TELOMY

SCIENCE MEETS YOU

Bionic Wristband

Design language & engineering specification

Researched dimensions, form rules and a parametric build plan, written so that Claude or a CAD designer can model the complete product in Autodesk Fusion. Every dimension in this document matches parameters_fusion.csv and master_spec.json in the same pack.

46 × 24
POD L × W (MM)

9.5 mm
MAX THICKNESS, INCL. 0.9 DOME

18 mm
STRAP WIDTH, QUICK-RELEASE

130–210
WRIST FIT S + L STRAPS (MM)



01 DESIGN LANGUAGE

Design language: the infinity, drawn in light

01 Light draws the infinity

The silhouette stays calm: a two-lobe stadium with only a 1 mm waist. The ∞ exists only as two warm light loops, lemniscate halves whose tails flank the dark central window, where the crossing is implied. When the band is idle, the light is off and the object is quiet.

02 One continuous skin

Every exterior transition is curvature-continuous (G2). There are no sharp edges; the minimum radius is 0.5. Shell, wings and strap read as one moulded organism, not a "pod plus strap".

03 The titanium horizon

One bright line only: a 1.2 mm polished Ti band around the pod side. It is structure, the top ECG electrode and the antenna break, all in one gesture.

04 Calm crown, sculpted shoulders

The top is near-flat where the electronics live. The sculpture moves to the 12 mm strap wings, which flare from 18 to 24 mm and start the wrap at R22.

05 Science, not decoration

No icons, no text on top. The ∞ mark (3 mm) is the only symbol; the wordmark lives on the caseback and inside the strap. Data is shown by light rhythm, not by screens.

06 Warm technology

Ivory ceramic, champagne light and soft-touch elastomer. It is premium and human; it should not look medical or like a gadget.



REFERENCE: TOP VIEW (ABOVE), LIGHT INDICATOR AND FRONT (BELOW)



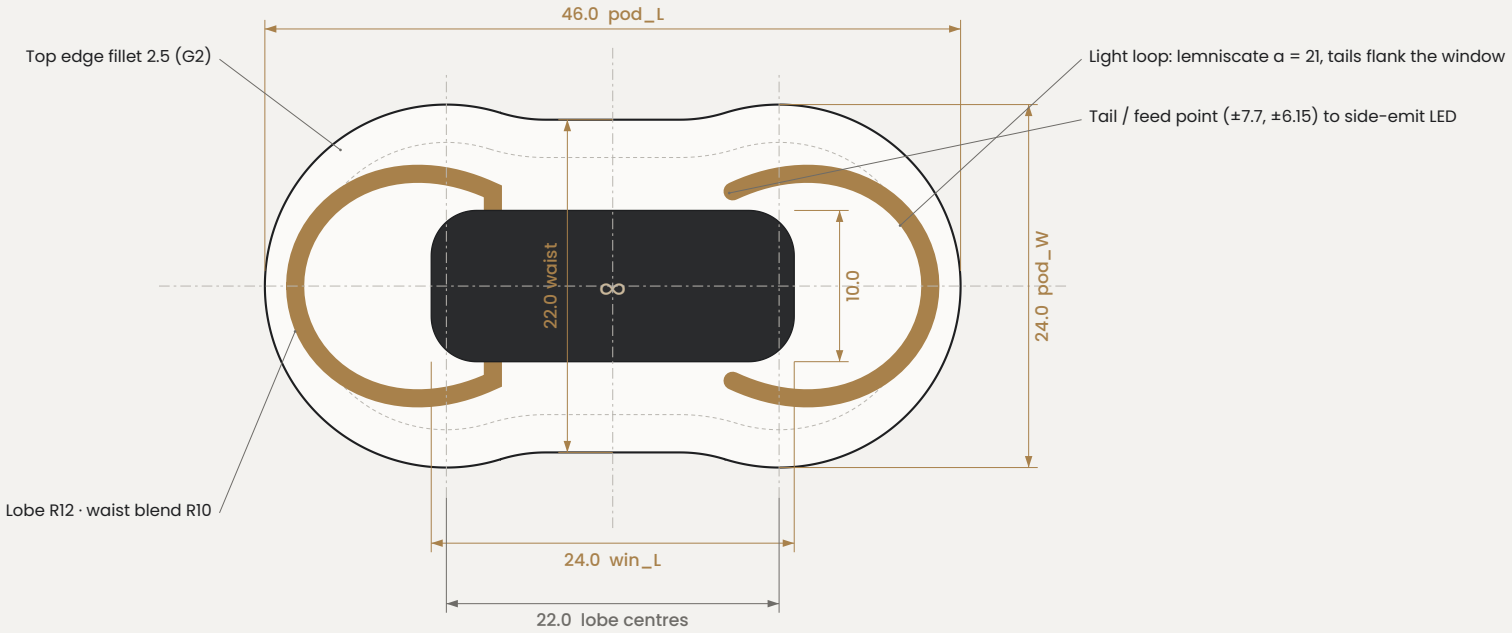


KEY PROPORTIONS

Ratio	Value	Rule
Pod length : width	46 : 24 = 1.92	Keep between 1.8 and 2.0
Visible top (pod + wings) : width	70 : 24 = 2.9	Long, calm, reads as a band, not a watch
Thickness : width	9.5 : 24 = 0.40	Under 0.45 so it never looks like a puck
Light line : pod width	1.2 : 24 = 1/20	A hairline. Never thicker than 1.5

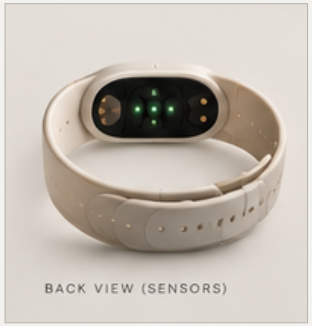
02 FORM GEOMETRY

Plan view: outline, window and light loops (2:1)



DRAWN AT 2:1. Light groove 1.2 W × 0.8 D, ≥ 2.0 from edge. Window: smoked PMMA/sapphire 0.8, R3 corners, flush to the crown.





Plan parameter	Value
pod_L × pod_W	46.0 × 24.0
pod_waist_W	22.0
lobe_R @ lobe_cx	R12 @ ±11
waist_blend_R	10.0
Plan area	950 mm²
win_L × win_W, win_R	24 × 10, R3
light_W × light_D	1.2 × 0.8

03 RESEARCH BASIS

Who it must fit: Indian wrist data and benchmarks

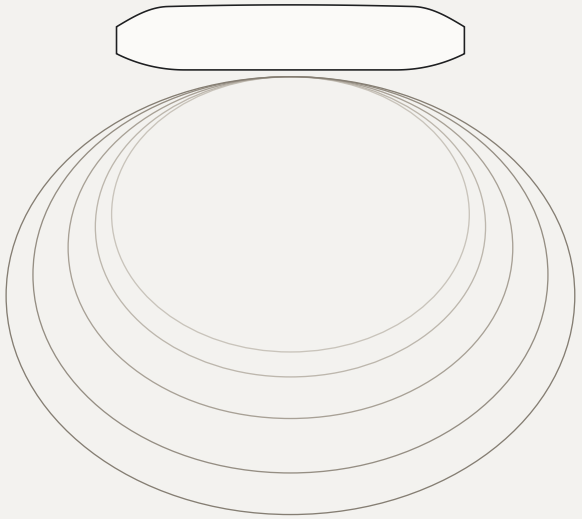
Measure	Value	Source / note
Wrist circumference, Indian men	163.5 ± 9.4 mm	Kerala hospital sample, n = 112 (type 2 diabetes cohort)
Wrist circumference, Indian women	157.0 ± 9.7 mm	Same Kerala sample (likely above general-population average)
Wrist circumference, Indian women	143.6 mm; P5 132, P95 155	Ahmedabad, n = 95 (Chakrabarti, hand anthropometry)
Wrist breadth, Indian women	46.1 mm; P5 41, P95 60	Same Ahmedabad study; smaller than US women (58.2)
Design range adopted	130–210 mm	S strap 130–172, L strap 154–210 (Xiaomi Band 9: 135–210)

BENCHMARKS (PUBLISHED SPECS)

Device	L × W × T (mm)	Pod mass	Battery	Life
WHOOP 4.0	35.97 × 25 × 10.1	11.3 g	192 mAh	≤ 5 d
Fitbit Charge 6	36.8 × 23.1 × 11.2	15 g	n/a	≤ 7 d
Xiaomi Smart Band 9	46.53 × 21.63 × 10.95	15.8 g	233 mAh	≤ 21 d
TELOMY v1	**46 × 24 × 9.5**	**≈ 15–16 g**	**≈ 140 mAh**	**4–5 d (est.)**

WHAT THIS MEANS FOR THE DESIGN

- **Thinnest in class at 9.5 mm**, but battery volume is the price. 7 days is a stretch target, not a promise (see page 7).
- Indian wrists are small. A rigid pod much longer than 46 mm overhangs on the P5 female wrist, so the length is capped there and the band gets its visual length from soft wings.
- **Wear position:** one finger-width above the ulnar styloid (wrist bone), snug. The PPG dome must keep contact without needing a tight strap.



Pod side profile on wrist phantoms, 1:1

Circumference 132 / 144 / 164 / 190 / 210 mm (breadth : depth = 1.3)

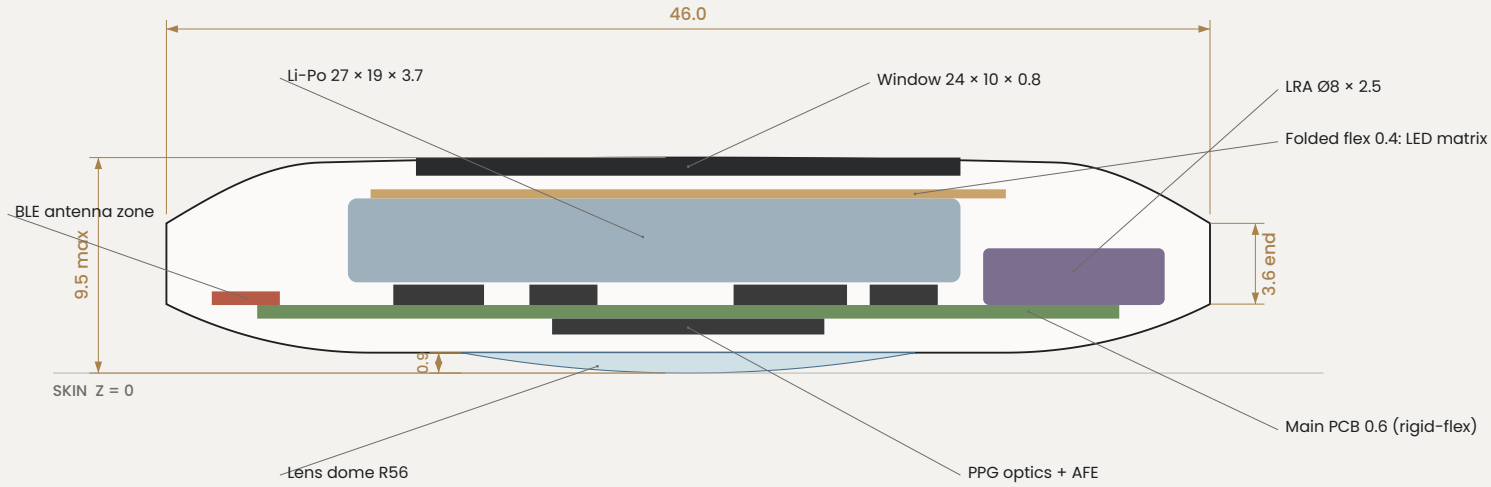
The caseback is flat for |X| ≤ 14 and then lifts at R20 ("ski tip"). The 0.9 mm dome guarantees PPG contact on flat, large wrists (R ≈ 45). On the smallest wrist (R ≈ 31) the flat edges float about 2 mm, which is acceptable, and the dome still carries the contact.





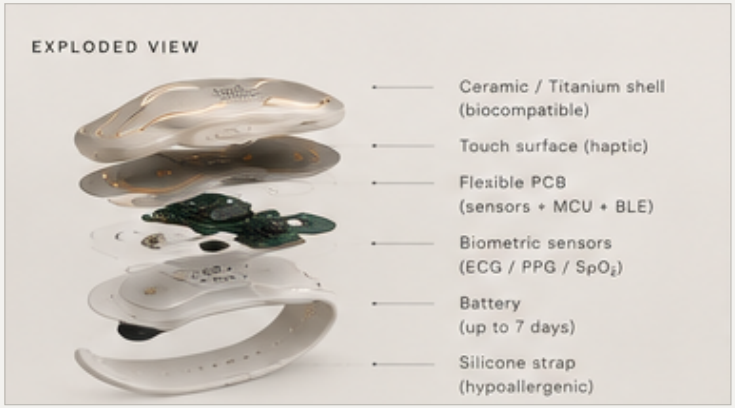
04 THICKNESS 9.5 MM

Side section and Z-stack (3:1)



LONGITUDINAL SECTION AT Y = 0, 3:1. Crown near-flat (R600) for |X| ≤ 16, falling to Z 6.6 at the ends; caseback flat for |X| ≤ 14, rising at R20.

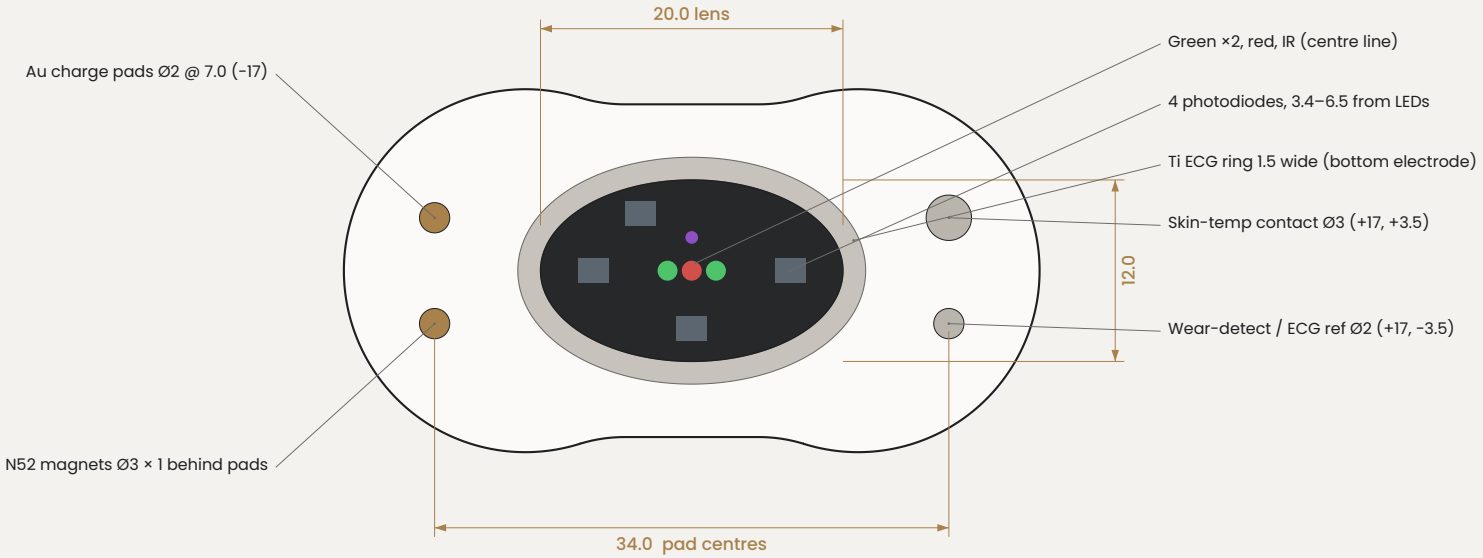
Layer (skin up)	Z from–to	T
Sensor dome	0.0–0.9	0.9
Caseback base	0.9–1.7	0.8
PPG optics / AFE (PCB bottom)	1.7–2.4	0.7
Main PCB	2.4–3.0	0.6
Parts under battery	3.0–4.0	1.0
Li-Po battery	4.0–7.7	3.7
Folded flex + LEDs	7.7–8.1	0.4
Swell allowance + gap	8.1–8.5	0.4
Top shell / window	8.5–9.5	1.0





05 SKIN INTERFACE

Caseback: sensors and contacts (2:1)



SENSING ARCHITECTURE

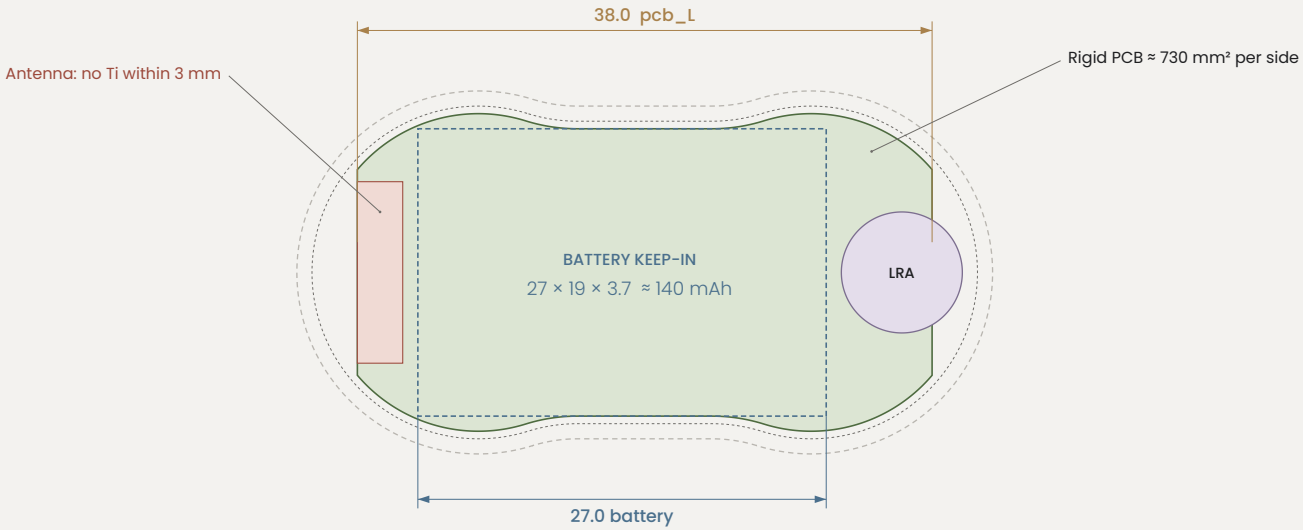
- **ECG (single lead):** the bottom Ti ring touches the wrist; the user pinches the Ti horizon line with the other hand. The two frame halves are split by a 0.6 mm insulating break. ECG is on demand (30 s), as on the Fitbit Charge 6, not continuous.
- **PPG / SpO2:** multi-wavelength (green ×2, red, IR) with 4 photodiodes and opaque barriers inside the lens. The dome rises 0.9 mm for contact without a tight strap; low-to-moderate contact pressure gives the best signal.
- **Temperature:** metal contact thermally isolated from the battery with foam, plus an internal reference sensor for compensation.
- **Charging:** 2-pin magnetic pogo cradle. No wireless charging, which keeps the 9.5 mm thickness.





06 ELECTRONICS PACKAGING

Internals: PCB envelope, battery, haptics (2:1)

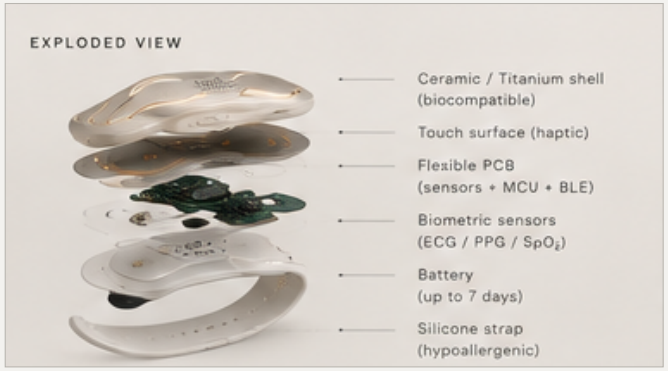


Block	Suggested part class	Where
SoC + BLE 5.3	nRF5340 / nRF54L WLCSP, 32 MHz xtal, DC-DC	PCB top, under battery (≤ 1.0 tall)
PPG / SpO2 AFE	MAX86141 or ADPD4100 class + 4 LEDs + 4 PDs	PCB bottom, centre under the lens
ECG AFE	MAX30003 / AD8233 class	PCB top; springs to the ring and frame halves
IMU	6-axis (BMI270 / LSM6DSO class)	PCB top
Temperature	MAX30208 / TMP117 + reference	Bottom contact at +17
Power	nPM1300-class PMIC, charger + fuel gauge	PCB top, -X end
Display + light	IS31FL3731-class driver, 15 × 6 LED matrix, 4 side-emit LEDs	Folded flex on top of battery
Haptics	Z-axis LRA Ø8 × 2.5 + driver	+X end beside battery

BATTERY VS THICKNESS: DECISION NEEDED

Option	Thickness	Battery	Est. life
Baseline	9.5 mm	27×19×3.7 ≈ 140 mAh	4–5 days
Long-life	10.5 mm	27×19×4.7 ≈ 180 mAh	6–7 days

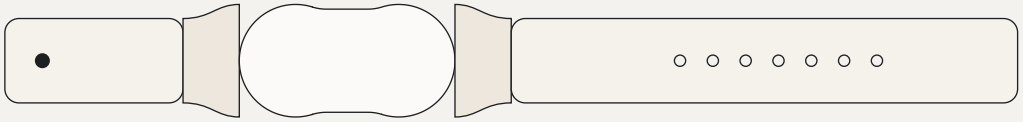
Life estimates assume continuous duty-cycled PPG, a few ECG checks a day, BLE batching and the light mostly off. WHOOP 4.0 gets ≤ 5 days from 192 mAh, so the "up to 7 days" on the concept board needs the long-life option or aggressive duty cycling. Validate on an EVT board.



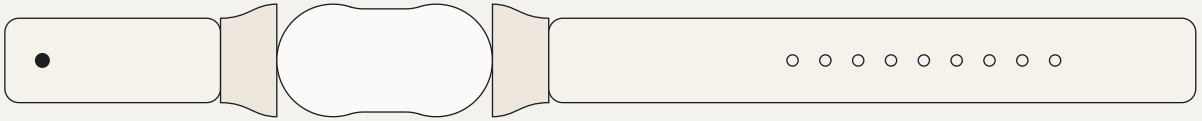
07 STRAP

Strap system: wings, quick-release and sizing

S STRAP · fits 130–172 mm · flat length 216 mm · 7 holes Ø2.4 @ 7.0



L STRAP · fits 154–210 mm · flat length 254 mm · 9 holes Ø2.4 @ 7.0



Drawn at 0.62:1. Hole distances measured from the wing root. Fit = 64 (pod + wings, inside) + stud distance + hole distance.

Parameter	Value
Strap width × thickness	18.0 × 2.8, edges R1.0
Material	FKM 70A (premium) / LSR 60A
Wing	12 long, 24 to 18 wide, G2 loft, wraps at R22
Attachment	18 mm quick-release spring bar, Ø1.5, tips Ø0.8
Lug	Hidden Ti horns, bar axis X ±24.5, Z 4.0
Closure	Pin-and-tuck stud: head Ø3.2, neck Ø2.4
S holes (from wing root)	36, 43, 50, 57, 64, 71, 78
L holes (from wing root)	52, 59, 66, 73, 80, 87, 94, 101, 108
Inner surface	Ventilation grooves 0.3 deep @ 1.0 pitch







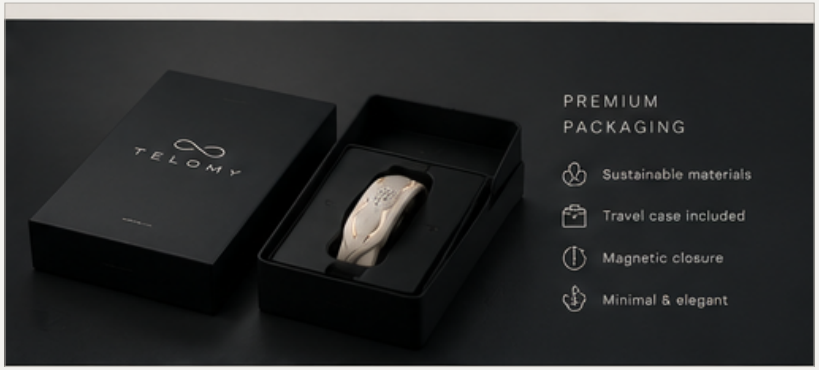
08 CMF

Colour, material and finish





Finish	Hex	Top shell	Strap	Light
Pearl White	#E8E4DE	Zirconia, white, satin polish	FKM ivory	2700 K
Midnight Grey	#3A3C3E	Zirconia, grey, matte	FKM graphite	6500 K
Champagne Gold	#C9AE92	Ti Gr2 + champagne PVD, bead-blast	FKM sand	2700 K
Forest Green	#5E6B61	Zirconia, green, matte	FKM forest	4000 K
Matte Black	#1A1A18	Zirconia, black, matte	FKM black	3000 K



MATERIAL NOTES

- Zirconia is RF-transparent, scratch-proof and warm to the touch (≈ 6.05 g/cm³). Keep walls 0.9–1.2 for ceramic injection moulding.
- The Ti Gr5 horizon line is polished on every colourway: the one constant bright detail across the range.
- FKM over silicone for a premium hand-feel and sweat and sunscreen resistance in Indian summers.
- Skin-contact parts (caseback, ring, strap) need ISO 10993-5/-10 testing.
- Hex values are sampled from the renders. Confirm with physical chips before tooling.

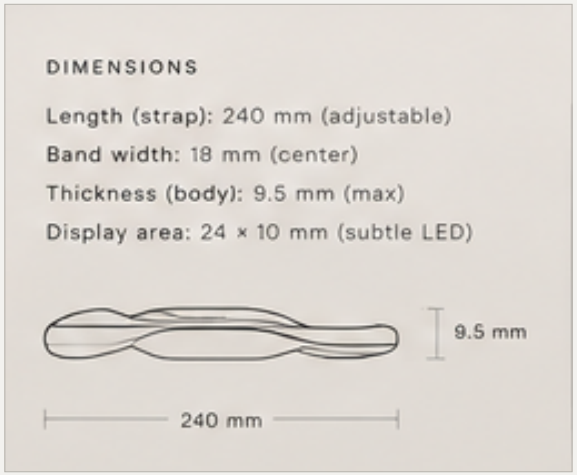
09 HOW TO USE THIS PACK

Building it in Fusion with Claude

File	Use
CLAUDE_FUSION_BUILD_BRIEF.md	Paste to Claude first: modelling sequence A–L, rules, checks, DFM
fusion_script/TelomyWristband/	Fusion script: 80 user parameters + master layout + keep-ins + phantoms
parameters_fusion.csv	The same parameters as a table (Parameter I/O import, or read by Claude)
master_spec.json	Machine-readable spec: parameters, profiles, Z-stack, materials, targets
dxf/telomy_wristband_views_1to1.dxf	Top, bottom, internals, side profile, fit check and strap flats, all 1:1
reference_images/	18 crops of the concept board (styling only)

STEPS

1. In Fusion: Utilities, then Add-Ins, then Scripts and Add-Ins, then +. Select the TelomyWristband folder and press Run. A step log appears.
2. Give Claude the build brief, the JSON and the step log. Ask it to continue from step B, "Pod master body". If Claude can drive Fusion (through a Fusion connector or by writing Fusion API scripts), it can then work feature by feature.
3. After each step, run its acceptance check (interference, wall thickness, zebra, phantom fit).
4. Decide the four open items (battery/thickness, display type, shell material, strap system) before CMF and drawings.



BOARD DIMENSIONS (240 / 18 / 9.5 / 24 X 10) ARE HONOURED OR EXPLAINED IN THIS SPEC

SOURCES

1. Wrist circumference in young adults with T2DM, Kerala (IJRMS, msjonline.org)
2. Chakrabarti et al., Hand anthropometry of Indian women (Ahmedabad)
3. Wrist circumference cut-offs, PMC11120788 (measurement landmark)
4. Xiaomi Smart Band 9 specifications (mi.com)
5. WHOOP 4.0 specifications (gadgetsandwearables.com)
6. Fitbit Charge 6 specifications (androidcentral.com)
7. Wrist PPG measurement site and applied pressure (literature search)
8. TELOMY concept board (supplied)

Limits: no reliable Indian wrist-depth data was found, so the phantoms use a 1.3 breadth : depth ratio. Validate with 20+ wrist casts or 3D scans.

